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A review of multimodal large language models for smart grids management and control

G. Cirrincione [b], M. Di Silvestre [a], R. Musca [a], Z. Naeem [a], E. Riva Sanseverino [a], G. Sciumè [a,*], G. Zizzo [a]

[a] Department of Engineering, University of Palermo, Viale delle Scienze, Building 9, Palermo, 90133, Italy
[b] Department of Engineering, Universitè de Picardie Jules Verne, VAvenue des Facultes, Le Bailly, Amiens, 90133, France

HIGHLIGHTS

- This review explores the role of multimodal LLMs in smart grid management for decision-making, fault diagnosis, and operational planning.
- Analysis of architectural and training aspects including the use of pretrained encoders, LoRA fine-tuning, and specialized loss functions.
- Practical considerations regarding the industrial implementation and integration of LLM models into the power systems environments.
- Multimodal LLMs for network resilience, renewable energy integration, and human–machine collaboration.

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ABSTRACT

The worldwide effort to reach carbon peak and neutrality objectives alongside energy market expansion has accelerated renewable energy integration, like wind and solar power. The shift towards renewable energy integration introduces substantial uncertainties in power system scheduling and control processes, which test the limits of existing theoretical methods. The advanced reasoning and data-processing capabilities of Large Language Models (LLMs), with particular reference to their ability to analyze multimodal data, provide transformative potential for managing and controlling smart grids. This review examines how LLMs can tackle modern power system challenges while confirming their fit with the power sector's expanding dependency on Artificial Intelligence (AI) technologies. We assess the requirements of modern power systems for such AI-based solutions, while evaluating how LLMs shape grid management and exploring their enabling technologies, such as model architecture and training methods, along with necessary data. Our review investigates how multimodal LLM technology serves different smart grid functions, including generation, transmission, distribution, consumption, and equipment management, to exhibit its adaptable nature in strengthening grid resilience and efficiency.

1. Introduction

The shift towards decarbonization and clean energy solutions has positioned wind and solar power as key players in the modernization of power systems. The essential role of renewable resources in reaching carbon peak and neutrality goals faces operational challenges because their unpredictable and intermittent nature makes grid management particularly difficult during operational scheduling and control optimization. Advances in sensor technology and digitalization have also set the framework for smart grids instantiation and evolution for proactive outage management and maintenance. Conventional approaches that rely on deterministic models and centralized optimization struggle to maintain a balance between grid stability, reliability, and economic efficiency, particularly in the face of variable renewable energy sources [1] and the integration of distributed assets such as energy storage and demand response programs. In contrast, distributed optimization methods offer superior scalability, but are hindered by their dependence on accurate forecasting, iterative computations, and rigid mathematical decoupling. These factors restrict their effectiveness in managing the high-dimensional, stochastic, and non-linear dynamics of modern power grids. In this frame, AI technologies, especially LLMs, stand out as revolutionary tools. They are at the base of modern Natural Language Processing (NLP) and the so-called 'Generative AI' (Gen-AI). The latter

* Corresponding author.
Email address: giuseppe.sciume01@unipa.it (G. Sciumè).

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Nomenclature

ACMCA	Auto Cross Modal Correlation Attention
AI	Artificial Intelligence
ANN	Artificial Neural Networks
BERT	Bidirectional Encoder Representations from Transformers
BLIP-2	Bootstrapping Language-Image Pre-training-2
CA	Cross-Attention
CNN	Convolutional Neural Networks
DL	Deep Learning
DNN	Deep Neural Networks
DDPG	Deep Deterministic Policy Gradient
DDQN	Double Deep Q-Networks
DQN	Deep Q-Networks
DBNs	Deep Belief Networks
DTs	Digital Twins
Gen-AI	Generative Artificial Intelligence
GANs	Generative Adversarial Networks
GNN	Graph Neural Networks
GPT	Generative Pre-trained Transformer
HITL	Human-In-The-Loop
ITC	Image-Text Contrastive
ITM	Image-Text Matching
KG	Knowledge Graph
LLM	Large Language Model
LM	Language Modeling
LSTM	Long Short-Term Memory
ML	Machine Learning
MLM	Masked Language Modeling
MHA	Multi-Head Attention
LoRA	Low-Rank Adaptation
NLP	Natural Language Processing
PEFT	Parameter Efficient Fine Tuning
RL	Reinforcement Learning
RLHF	Reinforcement Learning from Human Feedback
RNN	Recurrent Neural Networks
ReLU	Rectified Linear Unit
VAEs	Variational Autoencoders

is a subset of deep learning, which in turn is a particular machine learning technique. Gen-AI uses mostly statistical models as well as Artificial Neural Networks (ANNs), and can process labeled and unlabeled data using different training methods (supervised, unsupervised, and semi-supervised). While discriminative models are used for classification and prediction of labels for data points, generative models rely on a learned probability distribution of existing data to generate new data instances. The result of the processing of the Gen-AI models is either natural language, image, video, or audio. LLMs are one typology of Gen-AI, as their output is in the form of new combinations of text in the form of natural-sounding language [2,3]. When integrated with image or audio generation modules, LLMs can also have visual or sound output. LLMs surpass conventional model-based optimization and prior AI methods like reinforcement learning and specialized neural networks, like Convolutional Neural Networks (CNNs), by processing multimodal data types, including text, weather patterns, and load profiles through equipment telemetry and human inputs to create adaptive real-time decisions. As an example, Bidirectional Encoder Representations from Transformers (BERT) and, more recently, Generative Pre-trained Transformer-4 (GPT-4) demonstrate exceptional abilities in pattern recognition and semantic reasoning [4], thanks to their training on extensive datasets. Their capacity for scenario prediction allows them to effectively address modern power systems' uncertainty-driven challenges. The capacity to integrate textual, numerical, and temporal information enables these models to connect human cognitive tasks like contingency analysis and policy interpretation with computational optimization tasks such as generation scheduling and reserve allocation, thus pushing forward the development of human-machine hybrid intelligence. This review evaluates how LLMs can contribute to smart grids management and control by examining their transformative impact on operational processes throughout generation, transmission, distribution, and consumption areas. Our analysis in particular shows how LLMs, with their multimodal processing ability, can help solve some key challenges in current grid optimization practices as follows:

- Supporting grid operators to respond quickly to alarms [5], failures, and rapidly changing conditions. In these situations, LLMs can summarize large volumes of log files, sensor data, historical records, alerts, and forecasts. Specifically, LLMs can support grid operators in fault management by disturbance analysis [6] and DER tripping classification [7]. LLM can be game changers during emergency response during extreme weather events [8–14]. LLM can also support visual inspection of power transmission lines using text and images as inputs [15].
- Answering operator queries in natural language and providing virtual assistance to support humans in grid or plant management or with the administrative burden [16–20].
- Power grid embedding for Optimal Power Flow to query LLMs also with graph and tabular representation and development of tailored in-context learning and fine-tuning protocols for these models, [21–24].
- Managing simulations enhanced with domain-specific knowledge, reasoning capabilities, and handling of simulation parameters [25].
- Scheduling [26] that takes into account the uncertainty of renewable energy production and power demand by using forecasts [27–33] and semantic sensor interpretation [34,35].
- Real-time optimization of energy storage dispatch and electric vehicle recharge supports distributed resource coordination alongside demand response management [36].
- Cybersecurity threat detection in smart grids where LLM can handle both structured and unstructured data and provide alerts [37–39].

Incorporation of LLMs into grid operations enables utilities to shift from reactive, rule-based systems to proactive, data-driven models, which decrease operational expenses and enhance renewable curtailment rates while maintaining low-carbon mandate compliance. When multimodal data are considered, for some smart grids applications, the benefit that can be attained is more evident and better balances the computational effort that is needed to run LLMs. The review in [40] explicitly mentions that state-of-the-art LLMs require substantial computation power and memory, which can limit their applicability in low-latency, safety-critical grid tasks (Protective relaying, Contingency analysis, Load balancing, Event-driven decision support). Because these operations require milliseconds to seconds responsiveness, large models may be too slow to execute in real time.

Although other papers propose interesting reviews about the application of LLM to smart grids operation, they are mostly focused on text unimodal inputs. Interest in Multimodal LLMs is growing rapidly, but their application to smart grids remains relatively limited. A systematic search conducted on Scopus and ScienceDirect shows that, out of more than 12,000 papers on LLMs in the energy and engineering sectors, less than 1% explicitly address Multimodal LLM architectures for smart grid or power system applications. However, the number of

Table 1
Reviews considering LLM application to smart grids management and control.

Title	Yr	Applications considered	LLM architecture focus and data fusion
Large Language Models in Power Systems: Enhancing Control and Decision-Making [41]	2025	Control/optimization, decision support, knowledge management, cybersecurity; a practical Llama-3 [42] agent stabilizes voltages via tap-changer and DER actions	Transformer-based LLMs (GPT, BERT, Llama); implementation uses Llama-3 via Ollama. No encoder–decoder deep dive. No data fusion deep dive
Review on Application and Challenges of Large Language Models in Power Grids [43]	2025	Operator assistance, SCADA alarm parsing, visualization; generation-load forecasting; planning/dispatch/OPF; EV charging; fault detection/cybersecurity; converter/EDA design.	Transformer fundamentals (encoder–decoder, self-attention); survey of GPT-4/PaLM 2/BERT, multimodal assistants, hybrid LLM-GNN and physics-informed variants. No data fusion deep dive
Adaptable semantic interoperability in heterogeneous smart grids using Large Language Models [44]	2025	Smart grid interoperability, protocol conversion (SEP 2.1, IEC 61968, OpenADR), semantic correction/completion, message alignment	Fine-tuned LLaMA with RAG (LaBSE embeddings); clear architecture description including finetuning, LoRA, retrieval. Multimodal interoperability is considered only across heterogeneous protocols. No data fusion deep dive.
Artificial intelligence and machine learning for smart grids: foundational paradigms .... [45]	2025	Wide smart-grid domain: load forecasting, predictive maintenance, anomaly and cyber-attack detection, EV integration, demand-side management, digital twins.	LLMs within broader AI paradigm; no specific model described; focus on LLM role in DTs, generative reasoning, context-aware analytics. Discusses multimodal AI/ML ecosystems (IoT, sensors, cyber-physical DTs). Multimodality in LLMs is only indirectly implied.
AI Large Models for Power Systems: A Survey and Outlook [46]	2025	Wide smart-grid domain: load forecasting, wind/solar forecasting, fault detection, DTs, simulations, optimization, scenario generation, operator assistants, and reinforcement-learning-powered control.	High level explanation of Transformer architecture, including encoder–decoder, self-attention, cross-attention, PEFT, RAG, chain-of-thought, multimodal architecture (encoders, projectors, LLM backbone). High level presentation of MLLMs and their role for fault diagnosys in power-systems but no reference to specific operational scenarios.
Exploring the Emerging Role of LLMs in Smart Grid Cybersecurity [37]	2025	Focused specifically on cybersecurity use cases: attack detection, intrusion detection, vulnerability analysis, anomaly detection in IEC-61850 traffic, LLM-based security automation (RAG, KG integration, RLHF), and cyber-attack simulation.	High level explanation of LLM architectures. Focus on risks (hallucinations, adversarial use), and cybersecurity-specific performance. Mentions of heterogeneous cyber-physical data, logs, IoT data, communication packets (e.g., GOOSE/SV), but multimodality in LLMs is not a primary focus.
A Comprehensive Review on the Application of Large Language Models (LLMs) in Power Systems [40]	2025	Wide smart-grid domain: Control and management, forecasting, stability, data management; simulations, optimization, scenario generation, power electronics.	General transformer architecture, including LLM evolution. Self-attention and multi-head attention; LLM customization: prompt engineering, RAG (term/embedding/hybrid), PEFT (LoRA, adapters), fine-tuning strategies; Hybrid symbolic-LLM, RL-enhanced models, digital-twin integration. Multimodality is acknowledged but not developed as a core theme.
Our review		Wide smart-grid domain with a focus on fault detection and management, cyber-attack detection and semantic sensors interpretation for forecasting.	Focus on LLM and MLLM. Transformer fundamentals, Self attention and cross attention mechanisms for data fusion, full data fusion deepdive. Rigorous, technical presentation of MLLMs and cross-modal fusion in smart grids applications.

such studies has grown significantly since 2023, reflecting a clear shift from unimodal, text-centric approaches to the integration of Multimodal data.

Table 1 provides a comparison of existing reviews on the topic of LLM application to smart grids management and control. As it can be noticed, no review provides till now an explanation about how Multimodal LLMs actually work (architecturally and mathematically) for smart grids applications and how smart grids operation and management can benefit from multimodal LLM. This review, differently from others existing in the literature, highlights the ways in which multimodality can serve the tasks that are characteristic of smart grid operation and how it can bring significant improvement.

2. Review methodology

Fig. 1 illustrates the number of papers published over the past ten years on the topic of Large Language Models (LLMs).

Starting from 2022, the growth pattern transitions from a gradual linear increase to an almost vertical surge, culminating in a dramatic explosion during the 2024–2025 period, with more than 45,000 papers published in total. This sharp rise, carrying the number of publications from only a few hundred at the outset to more than twenty-two thousand today, reflects not merely a surge in documentation, but a substantive shift in the scientific community's research priorities toward this field. The unprecedented expansion of the literature over such a short time span poses significant challenges in terms of comprehensiveness, bias control, and reproducibility, thus motivating the adoption of a systematic review framework. For this reason, the present study adopts a systematic literature review methodology, reported in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [47], to ensure transparency and reproducibility of the selection process. PRISMA allows to create a thematic funnel that provides for the repeatability and verification of excluded studies through four main steps: identification, screening, eligibility, and inclusion. In addition, general principles for structuring and synthesizing review articles, as discussed by Glasziou [48], were followed to support a coherent and critical interpretation of the papers found. As anticipated in Section 1, the main questions the present review aims to answer are:

- Which challenges in smart grid management and control can be effectively addressed by multimodal LLMs?
- Which smart grid applications derive the greatest benefit from multimodal LLM-based solutions?
- How do multimodal LLMs compare with traditional and existing AI-based approaches in terms of performance and adaptability?
- What are the main practical and research challenges associated with deploying multimodal LLMs in real-world smart grid applications?

Based on these questions, the two keywords "Large Language Models" and "Smart Grids" were selected to search for relevant studies in the Scopus database. The first keyword was "Large Language Models" applied to the subject areas of "Engineering" and "Energy". The last search was conducted in Nov 2025 and 12,316 papers were found. Adding the keyword "Smart Grids", the number decreases to 48 documents.

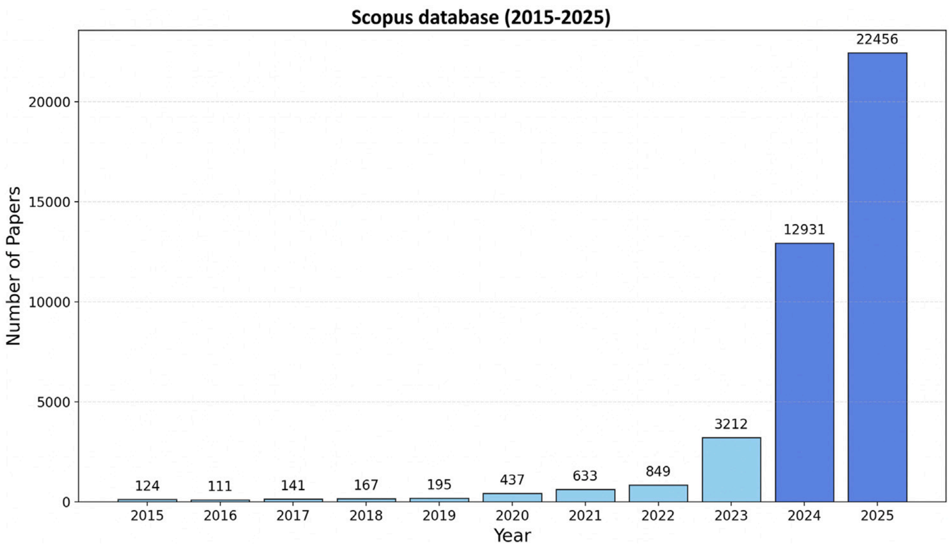

Fig. 1. Publication trends on LLM.

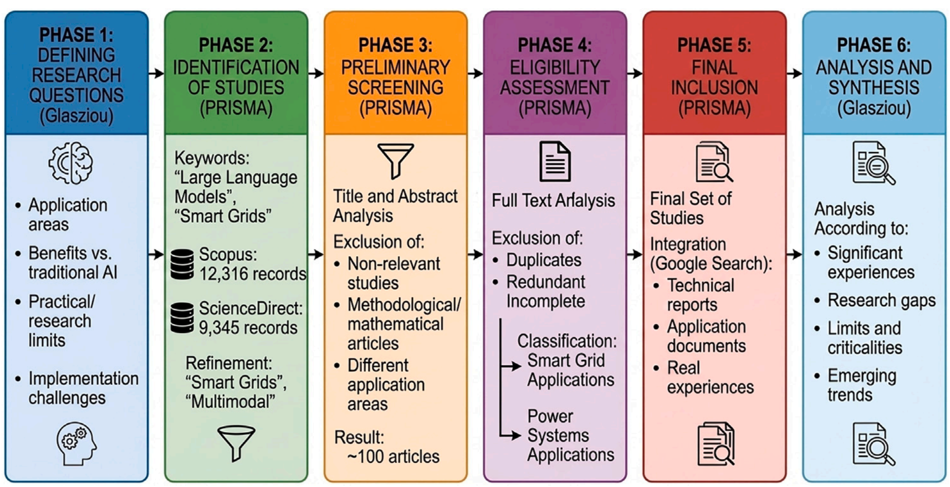

Fig. 2. Papers selection process flow-chart.

Similarly, a search of the ScienceDirect database using the two keywords and for the two identified subject areas produced 9345 results. Selecting the relevant publication titles (Applied Energy, Renewable and Sustainable Energy Reviews, Energy, International Journal of Electrical Power & Energy Systems, Energy Reports, Electric Power Systems Research) the number decreases to 1,592. To restrict the search field, "Multimodal" has been added to Large Language Models, and as expected, the number decreased to 57 papers. Those papers concerned different application fields, forecasting methods, methodology papers, mathematical models, review papers, etc. After an initial screening to exclude papers on other application fields, methodologies, and mathematical models, about 100 articles remained, which were categorized into LLM applications for Smart Grid or power systems. Finally, some of the 100 papers were discarded because they did not provide additional information with respect to other more complete papers in the list. In addition to the search on Scopus and ScienceDirect, reports and technical documents about implementation experiences were found using the Google search engine. All documents were analyzed covering, in particular, the following aspects:

(a) Most significant experiences in the field;
(b) Identification of gaps in the literature;
(c) Presentation and discussion on possible weaknesses identified by the authors;
(d) Identification of emerging trends and future research directions for the application of multimodal LLMs in smart grid/power system management and control.

The study selection process is summarized in the flow-chart showed in Fig. 2. According to the analysis criteria adopted, the manuscript is organized to provide a systematic analysis of points (a)–(d). The main application experiences of LLMs and multimodal models in electrical systems are discussed in Sections 3 and 4. The gaps in the literature and the limitations of existing approaches emerge from the comparison with traditional AI techniques and the critical analysis of current solutions. Methodological, architectural, and operational weaknesses are explored in depth in the sections dedicated to training (3.2 and 3.3), evaluation (3.4), and implementation requirements (3.5), while emerging trends and future research directions are summarized in Section 5.

3. Generative AI and large language models

In this section, the route leading to multimodal modern LLMs has been outlined, with a closer look at smart grids management and control.

The evolution of LLM applications in power systems can be traced through several distinct phases [40], beginning with the broader use of Artificial Intelligence (AI) in grid operations and culminating in modern, multimodal, agent-based LLM deployments. Before LLMs emerged, before 2020, power systems primarily relied on classic machine learning and deep learning approaches for forecasting, control, and fault detection. Between 2017 and 2021, the transformer architecture, was introduced by Vaswani et al. [49], providing the computational foundation later exploited in power system LLM applications. Models like GPT-2, GPT-3, and BERT, though not power system specific, showed unprecedented capabilities in text understanding and multi-step inference [50]. During this period, power systems research remained largely conventional, but the potential of transformers for grid applications began to be recognized. The first concrete step toward LLM integration into the power sector came via natural language interfaces, text classification, and document analysis [17,18], as well as prompt-learning-based extraction of emergency plan information [19]. Pretrained language models (e.g., multi-grained PLMs) were also proposed for audit-text classification [20]. These tools focused on knowledge access, documentation automation, and query handling, not on direct operational control. However, they mark the first explicit use of LLMs within the operational workflow of power utilities. By 2023–2024, LLM applications expanded significantly: GPT-enhanced Optimal Power Flow with linguistic constraints [24]; domain specific dispatch tools such as GAIA [26]; SCADA alarm processing using LLMs and knowledge graphs [5]; and forecasting and time-series modeling. Several studies introduced LLMs for forecasting tasks, such as wind power forecasting using fine-tuned BERT [27]; GPT-agent for probabilistic PV forecasting [28]; fine-tuning of a pretrained LLM to create bidding behavior agents, which forecast day-ahead bidding behaviors [29]; and LLM-driven load forecasting frameworks such as LFLLM and LFPLM [30,31]. Additionally, the multimodal Power-LLaVA, vision-language model for transmission line inspection was introduced [15]. Between 2024 and 2025 LLMs moved into agent-based control and simulation, with many examples from the Electric Vehicles area [51–54]. These works demonstrate a transition from simple NLP tasks to interactive, autonomous agents integrated with power system environments. In the most recent literature, LLMs have been considered for complex tasks involving grid protection, cybersecurity, and anomaly detection, such as LLM-enhanced cyberattack detection for smart inverters [39], LLM-based agents using step-by-step reasoning for disturbance analysis [6] and DER tripping classification using LLMs [7]. The latest works point toward multimodal and hybrid architectures and benchmarking [55]. As it will be later detailed, multimodality has significantly improved certain smart grids management and control applications, like as fault management, cybersecurity threat detection and semantic sensor reasoning. In the following sections, a deep dive into the rational of using multimodal data for smart grids applications is analyzed. First, the value of using multimodal data rather than unimodal data in several smart grids applications is considered, the way in which multimodal data are treated in LLMs is presented together with the description of loss functions for multimodal LLMs. Finally, the way to adapt LLMs to multimodal power industry applications is shown.

3.1. From machine learning to large language models for smart grids operation

Traditionally, different types of AI models are considered state-of-the-art for different types of data. For example CNNs are quite good at analyzing images as they imitate the way the human brain processes visual information. Coupled with Long Short-Term Memory (LSTM), which are in turn very good for time series analysis, they can provide a powerful tool for predictions. In [56], the authors describe a system that predicts residential energy consumption using LSTM and federated learning on edge devices. This approach protects privacy, reduces communication, and makes predictions more accurate (-8% error) and faster (-80% time) than centralized methods. Meanwhile, [57] describes a CNN-LSTM model that has been implemented for predicting photovoltaic power plant output based on sky photos and power output information.

Using CNN and LSTM has allowed the learning of spatial and temporal patterns from inputs, giving rise to higher performance compared to individual models and showing the potential of coupling images and numerical data. In [58] a review of papers using CNN for smart grid applications is presented. Most of them are related to energy theft detection, load disaggregation, and cyberattack detection (false data injection) [59]. In many cases, CNN is used in combination with other techniques, such as LSTM or autoencoders, again showing the potential of treating multiple data types together. The study in [60] proves that CNN works better for fault diagnosis if other input data are considered, else than measurements, such as contextual information like grid topology. Recurrent Neural Networks (RNNs), are also specialized architectures that, similarly to LSTM, have a feedback loop, allowing them to maintain a hidden state and process sequential data (e.g., text, time series) by considering past inputs. RNNs have also been used to solve classification problems, such as detecting the load margin at the network buses to prevent voltage instability in [61]. More frequently, RNNs are used to analyze text, but without much success, as contextual information can hardly be considered. Moreover, they cannot handle large sequences of text or data, like long paragraphs, essays, or long time series, and are subjected to what's called the vanishing/exploding gradient problem. Other issues arise with training time that cannot be accelerated by parallelization. In [62], the authors propose a Deep Learning-based method for identifying misconfigured network devices using only operational data, showing how the system can help operators improve network reliability. The paper [63] offers a comprehensive review about how Deep Learning, particularly Deep Neural Networks (DNNs), is used in power systems. It categorizes the use of DNNs into three main areas: discriminative models, generative models, and reinforcement learning approaches. Discriminative models are used for supervised learning tasks such as classification and regression. These include architectures like Rectified Linear Unit (ReLU) networks, Stacked Autoencoders (SAEs), LSTM networks, and CNNs. These models have been applied to a wide range of problems, including voltage stability assessment, load modeling, fault detection, and renewable energy forecasting. While they offer high generalization capabilities and fast inference, they often struggle with capturing temporal and spatial dependencies unless specifically designed to do so, as in the case of LSTM and CNN. Moreover, they cannot generally handle long data series. Generative models, such as Deep Belief Networks (DBNs), Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs), are used for probabilistic modeling and data synthesis. These models are particularly useful in scenarios involving uncertainty, such as state estimation, scenario generation for renewable energy, and synthetic data creation for power grid simulations. They are capable of learning complex data distributions but often require large datasets and can be computationally intensive to train. GANs, while powerful, may suffer from training instability and mode collapse, whereas VAEs offer more stable training but sometimes produce less sharp outputs. Reinforcement learning models, including Deep Q-Networks (DQN), Double DQNs (DDQN), and Deep Deterministic Policy Gradient (DDPG) methods, are employed for control and optimization tasks. These models learn optimal policies through interaction with the environment and have been successfully applied to voltage control, emergency response, energy scheduling, and cybersecurity in power systems. While they show promise in handling dynamic and uncertain environments, they can be slow to converge and sensitive to noise. When handling complex temporal and spatial patterns, and managing multimodal input data, all these models fail to provide effective solutions, for which mechanisms like 'attention' [49] in Transformer architectures, which weigh differently the importance of parts of the input string, have proved to be beneficial. Transfer learning and domain adaptation, aiming to improve model generalization across different datasets or operational contexts, reducing the need for extensive retraining, are some of the features that have also leveraged the success of Transformers.

In general, the main limitation of most of the approaches listed above is the ability to process efficiently only one input type (i.e., numerical data, time series data, images) and the need for retraining when tasks change slightly.

Transformers appear to be a big advancement in many aspects, as they can be parallelized, so that very large models can be easily trained. Moreover, they use self-attention or cross-attention mechanisms to process data (including multimodal data) in parallel, excelling at capturing long-range dependencies and achieving state-of-the-art results in many tasks. Finally, Transformers have a so-called ‘generative part’ that can be enriched with specialized modules dedicated to textual/visual/audio outputs so as to create LLM-based, very flexible ‘expert systems’.

3.2. Architecture and training of LLMs: unimodal versus multimodal data

The rapid progress of transformer-based LLMs represents a milestone in artificial intelligence [49,64], enabling machines not only to process text but also to integrate multimodal data such as images, audio, and numerical signals. While earlier machine learning models were effective within narrow domains, they struggled when data were heterogeneous or when long-range dependencies needed to be captured. Transformers addressed these limitations, and through their evolution into multimodal LLMs, they now offer promising capabilities for domains like smart grids, where operators must integrate telemetry, images, weather forecasts, and operational reports into a unified framework for monitoring and control. Modern grids produce massive heterogeneous data: time-series sensor signals, thermal images, tabular operational data, and unstructured text such as operator logs [65].

Unimodal Transformers, e.g., those trained solely on sensor time series, miss key contextual information, making them inadequate for high-fidelity grid intelligence.

At the heart of transformers lies the attention mechanism, which allows models to evaluate the importance of different elements in an input sequence. This innovation resolved several long-standing issues in deep learning: RNNs suffered from vanishing gradients and inability to capture long sequences, while CNNs were restricted to local receptive fields. Attention can be explicated in two forms: self-attention and cross-attention.

In transformers, each token in the input sequence is represented as a vector in a high-dimensional embedding space. Encoding is the process of assigning a number of real features to each of these tokens. The higher this number of features with embedding size d_k, the better description of that piece of information we can get. Assuming an input of size m, the encoded input will be $X \in \mathbb{R}^{m\times d_k}$. To process the encoded input, the model learns three squared matrices W_Q, W_K, W_V of size d_k that project X into Queries (Q), Keys (K), and Values (V):

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{1}$$

The self-attention operation is then:

$$SA(X) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{2}$$

The function $SA(X)$ produces a contextualized representation where each element incorporates information from all other elements within the sequence. The attention scores are computed through the matrix multiplication QK^T, which calculates the dot product between queries and the transpose of keys K^T, measuring the similarity between all pairs of positions in the sequence.

The scaling factor $\sqrt{d_k}$ serves as a normalization term that prevents attention scores from becoming excessively large as the embedding dimension increases, which could otherwise cause the softmax function to have extremely small gradients, hampering the learning process. The $softmax$ function converts these raw attention scores into a probability distribution where each row sums to unity, effectively determining the amount of attention allocated to each position in the sequence. Finally, this attention distribution weights the value matrix V to produce the contextualized output representation. This mechanism computes contextualized representations by weighting each token’s contribution based on similarity with others. In text, this captures dependencies across words; in images, relationships among patches; and in sensor data, correlations across time steps. In multimodal learning, cross-attention enables interaction between modalities. Also in this case, W_Q, W_K, and W_V are trainable weight matrices of size $d_k \times d_k$. Assume the encoded input Y being of a different modality compared to another encoded input X. Multiplying Y of the size $n \times d_k$ by W_Q produces the query matrix Q, which encodes “what the target modality is asking for”.

Multiplying X by W_K yields the key matrix K, which encodes “what information the source modality can offer”, while multiplying X by W_V produces the value matrix V, which contains the actual content to be passed on.

Given a target sequence Y (e.g., text) and a source sequence X (e.g., image patches), their embeddings are projected as:

$$Q' = YW_Q, \quad K = XW_K, \quad V = XW_V \tag{3}$$

The cross-attention operation is:

$$CA(Y, X) = \mathrm{softmax}\left(\frac{Q'K^\top}{\sqrt{d_k}}\right)V \tag{4}$$

Here, the matrix product $Q'K^T \in \mathbb{R}^{n\times m}$ computes a raw similarity score between each of the n query vectors and all m key vectors. Division by $\sqrt{d_k}$ serves to normalize these scores, preventing them from growing large in high-dimensional spaces and thereby avoiding vanishing gradients during backpropagation. The softmax function is applied row-wise to convert each row of similarity scores into a probability distribution over the m source positions, indicating how strongly each target element should attend to each source element. Finally, the attention weight matrix multiplies the value matrix $V \in \mathbb{R}^{m\times d_k}$ to produce an enriched representation of the target modality: each of the n output vectors is a weighted sum of the m source values. In this formulation, Y represents the target sequence or modality that seeks information, while X represents the source sequence or modality that provides information, creating an asymmetric relationship that enables directed information flow between different data streams [66]. Different fusion strategies are possible depending on the stage. In early fusion, modalities are concatenated at the input with the same embedding size and jointly processed. This provides holistic representations, but risks modality imbalance. In mid/late fusion, each modality undergoes self-attention separately. These contextualized representations are later fused via cross-attention or specialized fusion modules or simply compared (i.e., CLIP [67]). In bidirectional fusion, instead, queries can flow in both directions, e.g., text attending to images and images attending to text, enabling deeper alignment. Smart grid scenarios illustrate the tradeoffs. Early fusion may work when logs and measurements are tightly coupled, while mid/late fusion is preferable when modalities have distinct temporal or spatial structures, such as high-frequency telemetry vs. low-frequency operator notes. With the mid/late fusion strategy, self-attention is applied separately. By processing each modality independently first, the model can develop representations that are optimally suited for each data type without the complexity of cross-modal interactions interfering with this crucial intra-modal understanding phase. The resulting contextualized representations of the single modalities are then typically used in subsequent cross-attention layers where information can flow between modalities. This two-stage approach of first building strong intramodal representations and then enabling intermodal fusion has proven highly effective in multimodal applications such as image captioning, visual question answering, and cross-modal retrieval tasks [11]. This process enables the model to build a coherent understanding, in which local features can be contextualized within the global structure [68]. The bidirectional capability allows multimodal transformers to support reasoning tasks between different modalities, like converting one type of data into another, all within a single architectural framework [69]. These concepts can be applied to cross-attention late fusion for different types of input data, i.e., logs and electrical measurements:

$$Z^{\text{meas}\leftarrow\text{logs}} = \text{CA}\left(H^{\text{meas}}, H^{\text{logs}}\right) \tag{5}$$

where:

$$H^{\text{meas}} = \text{SA}\left(X^{\text{meas}}\right), \quad H^{\text{logs}} = \text{SA}\left(X^{\text{logs}}\right) \tag{6}$$

are derived from the first stage of independent single attention. The notation $Z_{\text{meas}\leftarrow\text{logs}}$ represents enhanced measurement representations (target) that have been enriched with contextual information extracted from log data (source). The directional arrow ← explicitly indicates the flow of information from logs to measurements. The contextualized representations H_{meas} and H_{logs} are derived from applying self-attention to their respective raw data modalities. The measurement subscript refers to numerical sensor data, telemetry readings, and electrical quantities commonly found in smart grid monitoring systems, while the logs subscript encompasses textual data, including maintenance records, operator notes, and historical fault reports [70,71].

Transformer-based multimodal models benefit from pretrained encoders. Pre-trained blocks are stacks of neural network layers and attention layers already trained with unsupervised learning algorithms, together with autoregressive generation [65] respectively during encoding and decoding. For example, LLAMA, BERT and GPT provide strong textual backbones; ResNet or Vision Transformers (ViTs) encode images; wav2vec handles audio. Early multimodal models trained everything end-to-end, but this proved computationally prohibitive. These early transformers used unimodal encoders to extract features that were then merged using concatenation or attention modules to create a unified representation space [72]. These models were trained end-to-end on image-text data and could not efficiently reuse pre-trained, single-modality models like BERT [73] for text or ResNet for images, see Fig. 3. This lack of adaptability made them inefficient for tasks like smart grid management. Modern approaches adopt modular designs: pretrained encoders are kept frozen (fixed weights), and lightweight bridging modules are introduced. These recent transformer-based multimodal models have evolved into unified architectures using a universal encoder [74], and are now capable of generative tasks as well [75]. A key development is the Bootstrapping Language-Image Pre-training-2 (BLIP-2) architecture [76], illustrated in Fig. 4, which employs a lightweight Q-Former module [77] to connect pre-trained image encoders and LLMs without the need for retraining them. This modularity allows reusing large pretrained models while adapting them to specific multimodal tasks without retraining the full architecture. In power systems, this enables for example leveraging pretrained NLP models to process technical reports while integrating them with telemetry encoders specialized for numerical data. Parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) further reduce computational overhead by updating only small subsets of parameters.

The multimodal LLM techniques highlighted in this review can be naturally interpreted within the broader framework of Foundation Models (FMs) [78]. In the FM paradigm, a model is first pre-trained through self-supervision on vast heterogeneous, unlabeled datasets, learning generalizable representations of system structure and dynamics, before being efficiently adapted to diverse downstream tasks through lightweight techniques such as parameter-efficient fine-tuning (e.g., LoRA). In smart grids applications, multimodal LLMs follow the same pattern: pretrained textual, visual, and time series encoders, often kept frozen, are integrated with transformer backbones via cross-attention projectors, enabling the model to unify sensor data, weather information, operational logs, and equipment imagery into a coherent latent space. This mirrors the FM idea of learning shared token-level abstractions across modalities. Moreover, domain knowledge is injected through KG, knowledge-graph, integration, physics-aligned losses, or human feedback driven alignment (Reinforcement Learning with Human Feedback, RLHF), analogous to the FM roadmap for introducing trust, verification, and domain constraints. As a result, the multimodal LLM becomes a domain-specialized FM for grid operations, capable of powering downstream applications such as semantic sensor interpretation, cybersecurity anomaly detection, or fault diagnosis, exactly as envisioned for GridFMs, where a single pretrained model serves as a versatile substrate from which many grid-relevant capabilities can be derived.

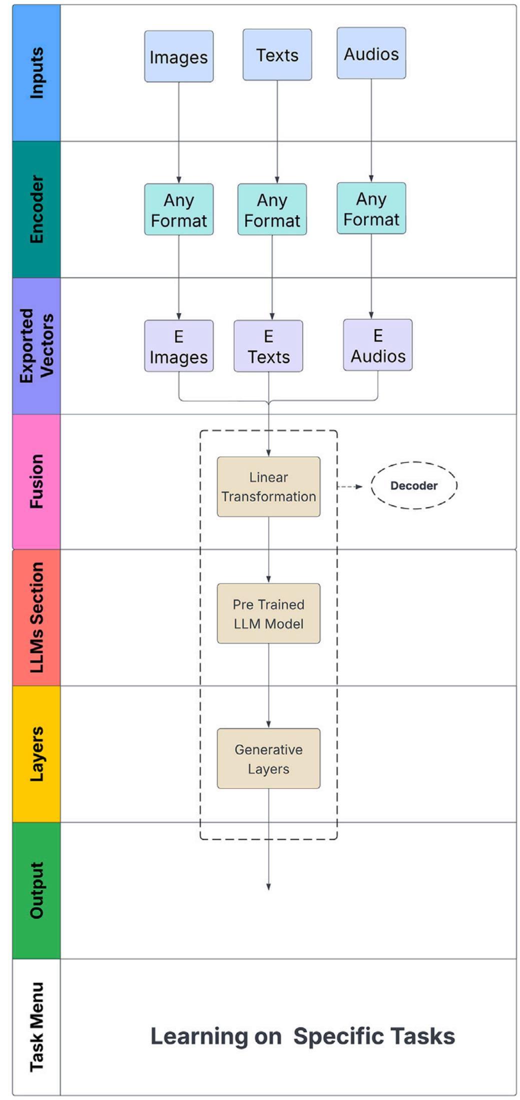

Fig. 3. Multimodal LLM: Functional Graph.

3.3. Loss functions for multimodal LLMs training

Architectures define potential, but loss functions determine how that potential is realized. It is like a measure of how well the model is performing during training [79]. Reducing the loss progressively brings the LLM to a better 'understanding' and language generation [80]. Such reduction is carried out during training, typically using gradient-based update rules on weights. Losses provide optimization signals that align modalities, enforce semantic and physical consistency and improve generative abilities. During the training phase of multimodal LLMs, four

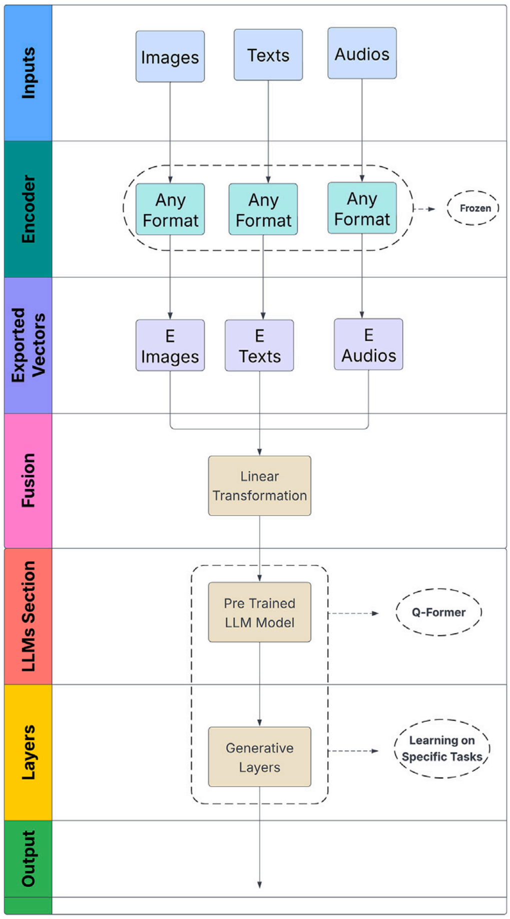

Fig. 4. BLIP-2 Multimodal Large Model: Functional Graph.

primary categories of loss functions can be used: Image-Text Contrastive (ITC) loss, Image-Text Matching (ITM) loss, Masked Language Modeling (MLM) loss, and Language Modeling (LM) loss. The foundation of ITC learning stems from the principles of contrastive learning [81]. The model processes many text-image tuples at a time and learns shared representations for images and texts by enhancing similarity (i.e., using dot product) between corresponding image-text pairs while reducing similarity between mismatched pairs. The purpose of ITM loss [82] is to assess how well images correspond to their associated text descriptions semantically. The method creates a binary classification task to optimize the relationship between matched image-text pairs. It processes one tuple at a time. The ITM loss prioritizes specific image-text pair alignment more strongly than ITC [67]. The MLM process [83] selects random words from the input sequence to mask and then demands the model to predict those words using surrounding context information. MLM encourages the model to derive semantic and syntactic context knowledge which leads to richer language representations. This self-supervised learning approach strengthens both the model's interpretive and generative abilities through valuable pre-trained features useful for multiple downstream NLP applications [84]. LM loss [85] aims to determine the probability distribution of upcoming words or characters based on surrounding context, including other modality features. The language modeling approach improves understanding and generation abilities by allowing models to learn both the probability distribution and structural patterns of language when they maximize next-word prediction accuracy. The model obtains basic language modeling and generation skills through this training process [86].

Loss functions can also include additive physics-informed terms or multimodal alignment terms, as discussed in Section 4.1. Learning refers to the process of updating the entries of the query, key, and value matrices, as well as all other trainable parameters described in the preceding sections. Parameter updates are performed using gradient-based optimization and the backpropagation algorithm. Learning can be end-to-end, when gradients propagate through the entire input-to-output pipeline, or partial, when only a subset of the architecture is trained. Partial training occurs, for example, when mask matrices, adapters, or frozen pretrained modules are used. Fine-tuning allows specialization of the model for sector-specific downstream tasks. Early fusion methods typically rely on end-to-end training because the modalities are fused early and must co-adapt. Late fusion methods, on the other hand, can benefit from frozen, pretrained encoders for each modality, with fusion layers trained on top. However, fine-tuning multimodal models often requires joint training of several or all components, depending on task complexity and architecture. Multimodal training strategies differ based on the fusion level at which the modalities are integrated. In Figs. 5 and 6, we illustrate training and fine-tuning workflows for Early Fusion (e.g., LLaVA [87]) and Late Fusion architectures. The parameters that go through update are indicated in bold and red, while the backpropagation for gradient correction is indicated starting from the loss and backpropagating to the encoders or to the projection matrices. Q, K and V are the query, key and values matrices and P is a trainable linear projection matrix. The embeddings of multimodal datasets, as well as the feedforward layers parameters, as well as other parameters that are not evidenced in the architecture, also may go through updates during learning. In Fig. 5, mi and mj are the token vectors used to calculate the loss function by dot product, softmax and logit. In Fig. 6, vi and tj are the token vectors referred to image and text respectively, used to calculate the loss function by dot product, softmax and logit. These schemas are only exemplary, as many possible training and fine-tuning methods are possible.

Multimodal LLMs therefore represent the convergence of powerful transformer architectures and carefully designed loss functions. Attention mechanisms, self-attention for intra-modal dependencies and cross-attention for inter-modal fusion, enable models to integrate text, images, and numerical signals. Pretrained encoders and modular designs reduce training costs, while contrastive, matching, and generative losses ensure alignment, semantic integrity, and fluency.

In the context of smart grids, these capabilities promise a paradigm shift. Operators can rely on systems that interpret multimodal evidence, generate intelligible explanations, and support robust, adaptive control. The path forward lies in domain-specific pretraining, physics-informed extensions, and human-in-the-loop feedback to ensure reliability in critical infrastructure.

3.4. Evaluation metrics for multimodal LLMs

(A 1.6) Multimodal capabilities of MLLMs, and their relevance to power systems engineering, lie in how these models interpret technical imagery, documentation, and structured constraints when paired with unstructured visual data. They show three key competency areas: multimodal recognition, perception, and reasoning, all of which expose strengths and limitations of current MLLMs in tasks analogous to those encountered in power engineering workflows [88]. The survey in [89] introduces a taxonomy of foundational, behavioral, and applied capabilities and highlights critical issues, such as data contamination, benchmark leakage, multimodal hallucinations, and insufficiently vision-centric tasks, that undermine classical LLM evaluation but become existential problems for MLLMs. It shows that metrics common in LLM evaluation are insufficient for multimodal tasks, where models may “answer correctly” for the wrong reasons or ignore visual content entirely. We focus here on multimodal reasoning, as it seems to be the most common task MLLMs are asked to perform in the smart grids area. A recent work [90] details different metrics to evaluate a model performance across multimodal reasoning tasks: Best Performance, Mean Relative Gain, Stability, and Adaptability. These

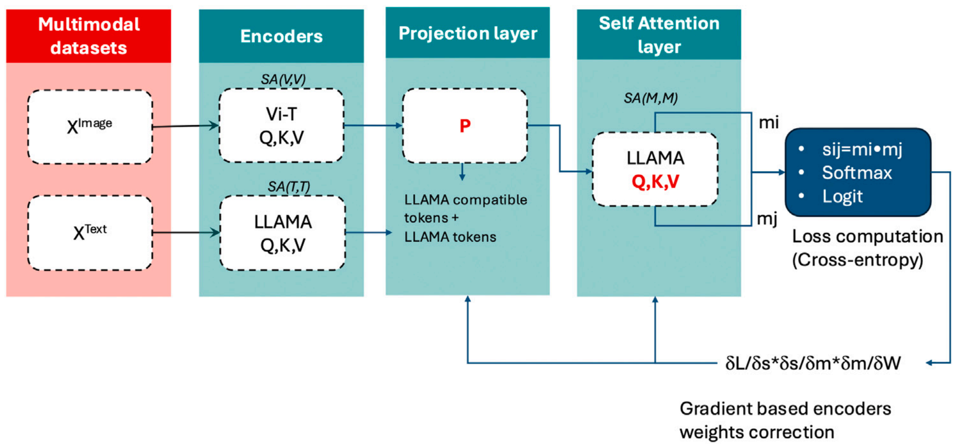

Fig. 5. Training and fine-tuning in early fusion, i.e., LLAVA.

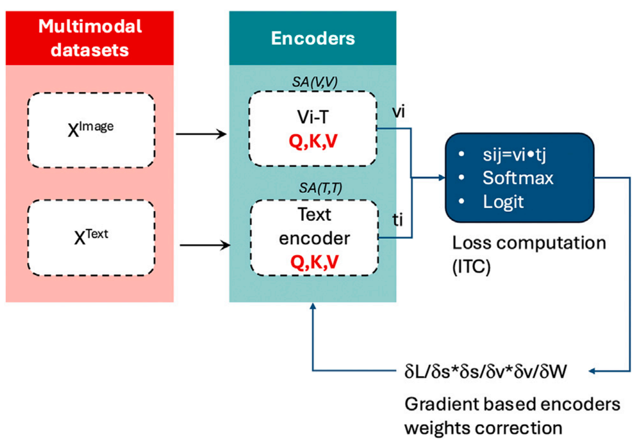

Fig. 6. Training and fine-tuning in late fusion, i.e., CLIP.

metrics compare different models $\mathcal{M}$, instructions $\mathcal{I}$, and datasets $\mathcal{D}$, starting from the definition of accuracy.

Accuracy

For model $m \in \mathcal{M}$, dataset $d \in \mathcal{D}$, and instruction $i \in \mathcal{I}$, accuracy acc_{mdi} is defined as:

$$\mathrm{acc}_{mdi} = \frac{1}{N_d} \sum_{j=1}^{N_d} \mathbf{1}\left(p^j_{mdi} = l^j_{mdi}\right), \mathrm{p}^{\mathrm{j}}_{\mathrm{mdi}} = \mathrm{LM}(\mathrm{T}^{\mathrm{j}}, \mathrm{V}^{\mathrm{j}}), \tag{7}$$

where LM represents a specific language model. The LLM takes only the text instruction, including text context, as input, denoted as T while the MLLM takes the multimodal context, (T,V) as input. In the above Eq. (7), N_d is the number of test instances in dataset d, T_j is the text context for instance j, V_j is the visual context for instance j, p^j_{mdi} is the model prediction, l^j_{mdi} is the ground–truth label and $\mathbf{1}(\cdot)$ is the indicator function.

Best performance

This metric identifies the maximum accuracy a model can reach across all instructions:

$$A_{md} = \max_{i \in \mathcal{I}} \left\{\mathrm{acc}_{mdi}\right\}, \qquad \tilde{i}_{md} = \arg\max_{i \in \mathcal{I}} \left\{\mathrm{acc}_{mdi}\right\}. \tag{8}$$

It measures the *upper capability limit* of a model on each dataset.

Mean relative gain (MRG)

This metric evaluates how much a model (or instruction) improves over the average baseline.

(a) MRG of models.

$$r_{mdi} = \frac{\mathrm{acc}_{mdi} - \mathrm{Acc}_{di}}{\mathrm{Acc}_{di}} \times 100, \tag{9}$$

$$\mathrm{Acc}_{di} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \mathrm{acc}_{mdi}, \tag{10}$$

$$\mathrm{MRG}^{\mathcal{M}}_{md} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} r_{mdi}. \tag{11}$$

This metric measures how much better or worse a model performs relative to the *average model* across instructions.

(b) MRG of instructions.

$$r_{idm} = \frac{\mathrm{acc}_{idm} - \mathrm{Acc}_{dm}}{\mathrm{Acc}_{dm}} \times 100, \tag{12}$$

$$\mathrm{Acc}_{dm} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathrm{acc}_{idm}, \tag{13}$$

$$\mathrm{MRG}^{\mathcal{I}}_{id} = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} r_{idm}. \tag{14}$$

This metric measures how effective an instruction is relative to the *average instruction* across models.

Stability

This quantifies how sensitive performance is to instruction changes (for a model) or model changes (for an instruction).

(a) model stability.

$$S'_{\mathcal{M},md} = \sqrt{\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \left(\mathrm{acc}_{mdi} - \mathrm{Acc}_{md}\right)^2}, \tag{15}$$

where Acc_{md} is the mean accuracy across instructions.

Lower values of this metric mean a model behaves *consistently* across different instructions.

(b) instruction stability.

$$S'_{\mathcal{I},id} = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \left(\mathrm{acc}_{idm} - \mathrm{Acc}_{id}\right)^2}. \tag{16}$$

Lower values of this metric mean an instruction performs *consistently* across different models.

Adaptability (Global top-k hit ratio)

This metric measures how frequently an instruction belongs to the top-K best-performing instructions for a given model:

$$I_{d,m}^{K} = \mathrm{Top}K\left(\{\mathrm{acc}_{mdi}\}_{i\in\mathcal{I}}\right), \tag{17}$$

$$\mathrm{GHR@}K_m = \frac{\mathrm{Counter}_{\mathcal{I},m}\left([I_{1,m}^{K}\| \dots \| I_{|\mathcal{D}|,m}^{K}]\right)}{|\mathcal{D}| \cdot K}, \tag{18}$$

where K is the number of top-performing instructions (typically $K = 3$), $\|$ is concatenation, Counter counts how many times each instruction appears in top-K. This metric measures how well instructions generalize across all datasets for a given model. Higher values mean an instruction is reliably strong.

Overall, while LLM evaluation focuses on reasoning and linguistic correctness, MLLM evaluation must jointly test perception, grounding, and cross-modal reasoning, requiring more sophisticated benchmarks, sample-specific criteria, and vision-aware evaluators that can accurately judge visual-textual alignment and reduce hallucination. In the next subsections, the features and the characteristics required to MLLMs for application to smart grid and electric power systems are discussed.

3.5. Requirements for applying MLLMs to smart grids industry applications

To effectively adapt LLMs for smart grid industry applications benefiting from multimodal data, the following technical pathway is recommended, [91]:

1. Multimodal data Collection and Preprocessing: gather and clean relevant datasets from the target industry/grid operator (domain adapted data) to ensure quality and consistency. Multimodal data in smart grids applications include textual documents like grid control policies and maintenance manuals, alongside sensor telemetry data like as voltage and frequency metrics, and operational logs containing historical fault records. As model performance depends heavily on data quality and suitability, they require preprocessing (i.e., data cleaning combined with denoising and annotation [92,93]).
2. Industry-Specific Knowledge Construction: build domain-specific knowledge bases or ontologies to enrich the model's contextual understanding, typically referring to the power system weather data or power devices. Standard approaches encompass manual data annotation techniques along with KG building and expert knowledge extraction processes [94]. In the power systems area, domain-specific expertise is integrated into the model. The framework can build KGs that display hierarchical relationships between components of the power system, such as transformers and substations, while embedding expert rules, such as voltage limits and safety protocols in direct decision-making processes.
3. Model Selection and Fine-Tuning: Choose an appropriate LLM architecture (encoder-decoder model, autoregressive model, autoencoder model), choose the fusion method (early, mid/late) and fine-tune it using industry-relevant data to specialize its capabilities.
4. Intelligent Task Decomposition and Execution: Break down complex tasks into manageable subtasks and automate their execution using the model [95].
5. Integration and Deployment: Implement the model within operational environments using agent-based frameworks such as AutoGPT [96] or similar systems.

Among these, the phase of model selection and fine-tuning is quite relevant. This phase typically employs pre-trained models like GPT and BERT that are adjusted through industry-specific data for specific tasks and situations, but also human expert feedback. To perform fine-tuning, hyperparameters and training set sampling strategies need to be changed, as well as the optimization algorithms. Lightweight fine-tuning methods like LoRA and parameter-efficient transfer learning help reduce computational load while improving downstream task performance [97]. As an example, an LLM fine-tuned with historical wind turbine fault diagnosis records can produce concrete repair suggestions when analyzing fresh sensor data. The study in [16] highlights how domain-specific training methods, including Reinforcement Learning with Human Feedback, RLHF, help fine-tune model outputs to match the needs of grid operators and their operational standards. The approach utilizes RL techniques that incorporate human feedback while applying multimodal KG methodologies to draw on domain experts' insights to tailor the model to industry specifications. The above workflow reaches its endpoint with integration and deployment through which the fine-tuned LLM finds its place in industrial systems using platforms like AutoGPT. This stage focuses on achieving smooth integration with existing legacy systems such as SCADA and energy management systems alongside comprehensive testing to assess performance metrics, including latency, scalability, and reliability.

4. Applications of multimodal LLMs in power systems

Although the literature still reports a small number of applications of LLMs and multimodal LLM to power systems, this number is steeply increasing. The works in [46,98] show a list of potential applications of LLM to the smart grids sector. However, not all of them need or make use of multimodal inputs. Out of the 30 cited applications, only a few make use of truly multimodal inputs, these being either data of different natures (i.e., text, images) or structured/unstructured data. Unlike natural language processing applications (i.e., for human machine interfacing), truly multimodal applications benefit the most from the use of LLM, thanks to the different possible data fusion mechanisms see Table 2.

As an example of a truly multimodal application for power systems, the smart power system agent described in [102] could autonomously collect data from grid sensors and smart meters, analyze these data to detect signs of equipment wear or potential faults, and then make recommendations or directly schedule maintenance activities before failures occur. At the same time, it could forecast demand changes based on historical trends and weather data, automatically adjust power distribution to balance supply and demand, and optimize integration of renewable energy sources. Through these autonomous and iterative actions, the system can help grid operators improve grid reliability, reduce operational costs, and enhance overall energy efficiency with minimal human intervention. When working with a single input modality, specialized algorithms can yield highly effective results. However, when multimodal data are available, integrating different types of inputs unlocks new potential—much like the human brain, which naturally combines diverse sensory information to enhance understanding. Multimodal inputs show different features. They can be discrete or continuous. In the power systems area, discrete data are typically related to images and text from a report or categories (i.e., day, week, month, season). While continuous data may relate to measurements of electrical quantities. Inputs may have different frequencies or densities. In the power systems area, measurements typically show a predictable frequency, related to the sampling frequency of the measurement device and the capacity of the TLC channel. Logs also have a predictable frequency, while images or other inputs, like reports or alarms, instead can be more frequent as they are generated on purpose during an event. Another important feature of a given data modality is the temporal or spatial structure. Images have a spatial structure, while measurements/time series have a temporal structure. However, in power systems, where the location in the grid is important to understand the 'context' and value of the measure, spatial structure can also be relevant (i.e., Phasor Measurement Units or other distributed measurement devices used collectively).

In the following subsections, three of the most relevant application fields from Table 2 will be discussed in more detail, focusing on Fault

Table 2
Multimodal data applications in power systems.

Application	Multimodal data description	Benefits of multimodal over unimodal data
Fault Localization and Repair [8–15]	Combines sensor readings, technician reports, equipment images, and maintenance logs	Enables faster and more accurate fault diagnosis and repair planning by correlating visual and textual evidence
Cybersecurity Threat Detection [37,38]	Integrates network traffic logs, system alerts, and threat intelligence reports in natural language	Detects complex, multi-stage attacks with better context and reduces false positives through semantic understanding
Digital Twin Creation [45]	Combines CAD drawings, sensor data, drone imagery, and maintenance records	Automates and updates virtual grid models with high fidelity, improving simulation accuracy and operational planning
Semantic Sensor Interpretation [32–35].	Links numerical sensor readings with metadata, historical trends, and equipment specs, including images	Provides contextualized, actionable insights rather than raw values, improving operator decision-making
Environmental Justice Analysis [99]	Combines demographic data, outage records, pollution maps, and infrastructure locations	Identifies disparities in grid reliability and environmental burden, supporting equitable policy decisions
Emergency Response Coordination [16]	Uses multilingual text, voice messages, and geospatial data	Ensures inclusive communication during crises, improving public safety and response effectiveness
Prosumer Behavior Modeling in RL [100,101]	Integrates consumption data, survey responses, market signals, and behavioral studies	Simulates realistic human responses in RL training, improving agent adaptability and reducing modeling bias
Scenario Generation [98, 102]	Uses historical grid data, expert narratives, and simulation parameters	Creates diverse and plausible training environments for classifiers or RL agents, enhancing robustness and preparedness

Localization and Repair, Cybersecurity Threat Detection, and Semantic Sensor Interpretation.

4.1. Fault localization and repair

As seen in Table 2, Fault Localization and Repair is one of the LLM applications where multi-modality can bring a great enhancement. In this area, the study in [13] compares four different advanced unimodal-fed methods (Support Vector Machine, Random Forest, LSTM, and CNN) with a multimodal early fusion Transformer based approach showing the latter outperforms all the formers by combining information from sensor time-series, thermal imaging and text fault logs. The paper employs late fusion at the feature-interaction level, since each modality—sensor time-series, thermal imaging, and text fault logs—is first processed independently through CNNs and bidirectional LSTM, and only then aligned and integrated through a multi-level cross-modal Transformer that performs adaptive feature-level alignment, multi-head self-attention, cross-attention, and final fusion with dynamic weighting rather than fusing modalities at the raw-data stage. In terms of computational complexity, the system is relatively heavy due to multi-branch feature extractors, multi-level Transformer attention layers, and a two-stage training scheme with cross-distillation, all of which increase training cost; however, once trained, it achieves faster inference than traditional multimodal fusion baselines thanks to its optimized alignment and attention mechanisms, as shown by its superior accuracy and lower processing time in comparative experiments. In [9], a sector-specific multimodal LLM integrating the semantic understanding capabilities of multimodal LLMs with historical fault data and operational procedures is proposed. Unlike traditional fault diagnosis methods that primarily handle structured numerical data (e.g., sensor readings and fault codes), the proposed LLM-based model can handle online and offline data. These include unstructured data such as text, natural language, images and data extracted via web scraping and application programming interfaces, as well as structured numerical data. All these data are converted into embeddings using specialized encoders. To improve task management capabilities while maintaining decision control, RL techniques based on expert feedback can be integrated. Fig. 7 illustrates a functional diagram of a generalized multimodal LLM specialized for fault diagnosis in power systems, where the expert feedback is part of the main architecture.

The arrows in Fig. 7 depict the flow and interaction of information between modules. Arrows from the multimodal dataset (Text, Video, Audio, Image) to “Integrated multimodal encoding” indicate input of diverse data types into the encoding module. The latter is generally constituted of a set of specialized encoders, one for each data type. The arrow from “Integrated multimodal encoding” to the LLM shows the transfer of encoded information into the model. Arrows between “System simulation” and the LLM add physics-based support to the learning phase, while those between “User collaboration” and the LLM represent communication with human experts.

The last phase employs a data-fusion mechanism that merges machine learning insights from the LLM with established physics-based principles to enhance diagnostic accuracy and reliability. The fusion happens by injecting physics-based constraints and simulator outputs into the attention-based fusion pipeline as additional tokens, masks, and losses, so that cross-attention weights preferentially aggregate source values that are consistent with power-system laws and digital-twin surrogates; mathematically, this augments the standard CA and MHA blocks (multiple parallel attention stacks) and adds physics-informed penalties to training [10].

In practice, simulators or physics surrogates produce features like state residuals, sensitivities, or stability margins that are aligned in time and equipment-space with language, logs, and images, then injected into cross-attention so target queries attend more to source evidence that also satisfies physics [103]. Training adds physics-informed losses alongside task losses, such as ITC, ITM or MLM. The network minimizes diagnostic loss while penalizing violations of power laws or stability margins computed on outputs or intermediate states, giving rise to a standard physics-informed ML pattern adapted to attention architectures. Eq. (19) shows the overall loss function.

$$\mathcal{L} = \mathcal{L}_{\text{diag}} + \lambda_{\text{phys}}\,\mathcal{L}_{\text{phys}} + \lambda_{\text{align}}\,\mathcal{L}_{\text{align}} \tag{19}$$

Here, the total loss $\mathcal{L}$ represents the overall objective function minimized during the training process. The diagnostic task loss $\mathcal{L}_{\text{diag}}$ corresponds to standard supervised learning objectives for specific applications such as fault classification, load forecasting, or equipment condition assessment.

The physics-informed loss component $\mathcal{L}_{\text{phys}}$ enforces compliance with physical laws and operational constraints inherent to power systems. This includes power flow equations of the form $P = VI\cos(\phi)$, voltage operating limits $V_{\min} \le V \le V_{\max}$, thermal constraints $I \le I_{\max}$, and various stability margins and power balance requirements. The multimodal alignment loss $\mathcal{L}_{\text{align}}$ ensures semantic consistency

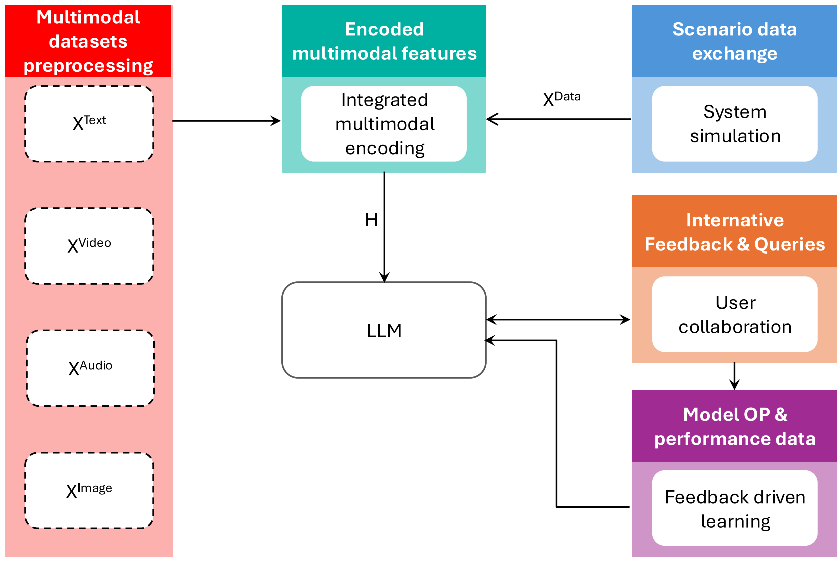

Fig. 7. Application of LLM in fault diagnosis of equipment.

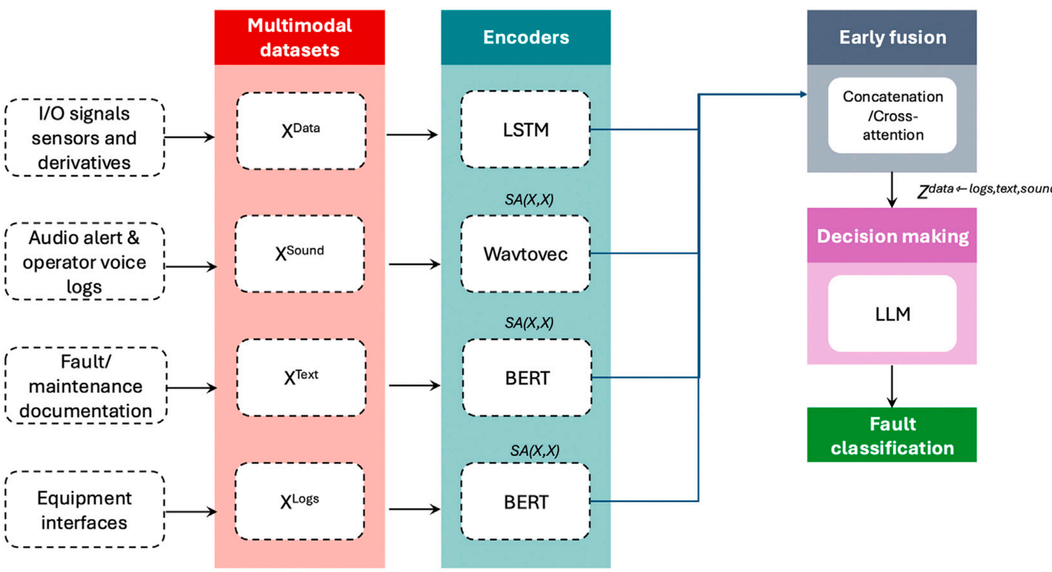

Fig. 8. Early fusion in fault localization and repair.

and proper alignment between different data modalities, incorporating image-text contrastive loss, image-text matching loss, and masked language modeling objectives.

The weighting parameters λ_{phys} and λ_{align} are non-negative hyperparameters that control the relative importance of physical constraint satisfaction and multimodal alignment, respectively, compared to the primary diagnostic task performance. This additive structure allows for careful balancing between task accuracy, physical realism, and multimodal coherence throughout the training process, ensuring that the learned model respects both data-driven patterns and domain-specific physical principles.

A similar methodology to the one just outlined is adopted in [11], where a Vision Learning Pretraining algorithm is used to develop a power-specific Vision Foundational Model. The goal is to match UAV-captured images with fault events on power transmission lines. Pretraining and task-specific training are conducted using ITC learning, which supports semantic alignment between visual and textual modalities—essential for defect recognition.

In such fault diagnosis applications, data fusion can be implemented both at an early and at a later stage. While in [9,12,13,15], data fusion is performed as early fusion at the features level, as detailed in Fig. 8, in [8,14], data fusion is performed at a later stage, during decision making, as detailed in Fig. 9. In early fusion, data are suitably encoded using pre-trained modules and are then projected onto the same features space, while in late fusion, data and other multimodal inputs are encoded separately and then used either to create prompts that are then fed to the LLM or to evaluate a similarity score, based on which the loss function can be calculated.

In the broader area of fault diagnosis and other operational functions within power systems, multimodal human-machine interfaces [104] are also employed. These interfaces enable operators to validate data and provide feedback, fostering a collaborative diagnostic ecosystem.

Other studies [8,14] use pre-trained LLMs in fault diagnosis by converting datasets into natural language prompts and applying supervised fine-tuning. In the initial training phase, each sample from the original dataset is transformed into prompt data using predefined templates.

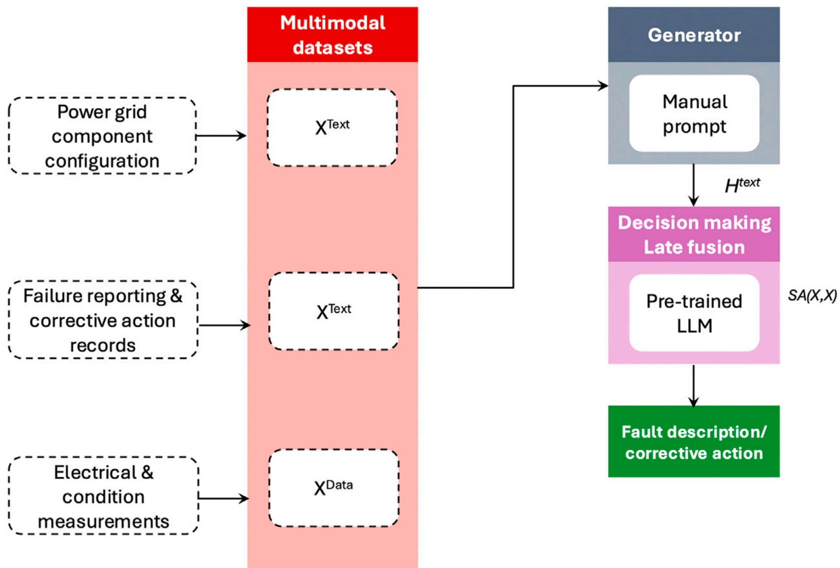

Fig. 9. Late fusion in fault localization and repair.

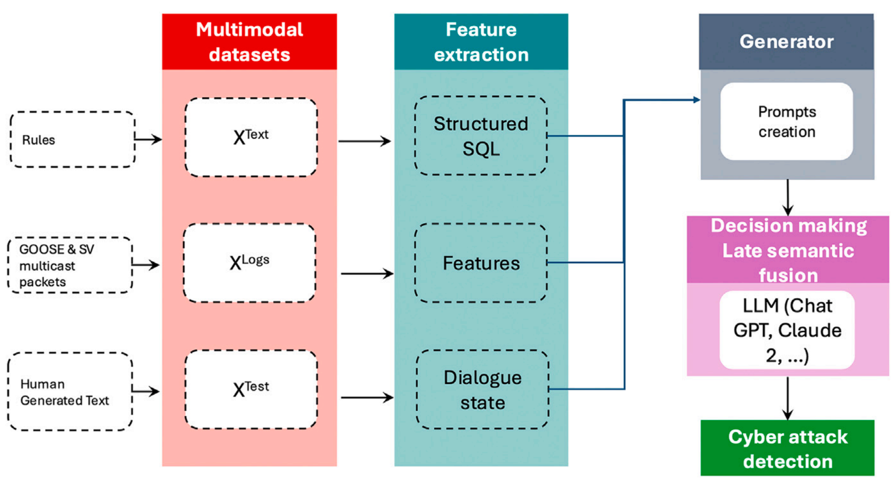

Fig. 10. Semantic late fusion in cybersecurity threat detection.

These prompts include contextual instructions related to the diagnosis issue, language monitoring information, and labeled responses. The resulting language-converted dataset is used to fine-tune LLMs without altering the model architecture or loss function. Both open-source (e.g., LLaMa-2) and closed-source (e.g., GPT-3.5-Turbo) LLMs are employed, with fine-tuning conducted via Low-Rank Adaptation (LoRA) and OpenAI's API playground.

4.2. Cybersecurity threats detection

Cybersecurity threat detection in smart grids is another area in which multimodality is quite effective. In this case, the latter refers to the combination of structured and unstructured data, rather than of data of different types, such as it happens in fault localization and repair.

In [37], a comprehensive review of cybersecurity threats in smart grids is presented. The paper evaluates the potential of LLMs to enhance protection, detection, and response. The layered architecture of smart grids and the vulnerabilities introduced by their reliance on digital communication and IoT technologies are presented. The paper then surveys existing cybersecurity techniques, including machine learning and deep learning approaches for intrusion detection, anomaly detection, and mitigation. It highlights the limitations of current models, such as their dependence on labeled data, lack of adaptability, and poor interpretability. In this context, LLMs are proposed as promising tools due to their ability to process heterogeneous data, understand context, and support automated decision-making. The paper outlines applications in which multimodal intrusion detection systems combine cyber and physical data—particularly in the context of Graph Neural Networks (GNNs). These systems do not involve transformer-based fusion strategies or attention mechanisms that operate across modalities.

The fusion of data types, where it occurs, is handled externally through preprocessing or model design (e.g., GNNs), not through integrated cross-modal attention layers.

In [38] the use of LLM, particularly ChatGPT 4.0, Claude 2, and Google Bard/PaLM 2, for cybersecurity in digital substations is explored. The LLM based system is used to detect anomalies in IEC 61,850-based communication protocols, specifically GOOSE and SV multicast messages, which are vulnerable to cyberattacks such as replay, injection, and denial-of-service. The paper describes a large-language-model-based Task-Oriented Dialogue (ToD) system that fuses heterogeneous data sources at the semantic level (GOOSE/SV communication features, human recommendations, SQL-based rules), as shown in Fig. 10. This

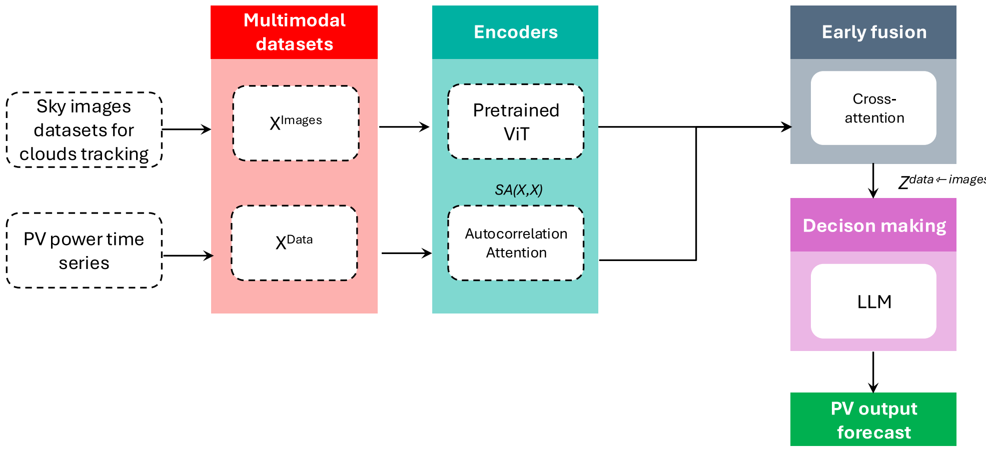

Fig. 11. Early fusion in semantic sensor interpretation.

highlights the LLM's ability to integrate: unstructured natural-language instructions, structured network data, and system-state descriptions. Thus the LLM creates a multimodal semantic understanding of the cyber-physical context. Traditional machine learning methods used for intrusion detection need retraining when new attack patterns emerge, which is resource-intensive and leaves systems exposed during adaptation gaps. In contrast, LLMs offer contextual understanding and adaptability, enabling them to detect novel threats without retraining. The authors propose a Human-In-The-Loop (HITL) framework where human experts guide the training and validation of LLMs. This approach improves model efficiency and reduces the need for retraining by incorporating expert recommendations into the preprocessing of communication data. The study uses a hardware-in-the-loop testbed to generate realistic datasets and evaluates the performance of different LLMs across different training levels. ChatGPT 4.0 consistently outperforms the other models in precision, recall, and F1-score, demonstrating its effectiveness in anomaly detection. Multimodal data use is present in the form of structured network communication logs (GOOSE and SV packets) combined with human-generated recommendations and textual representations of intrusion detection rules. However, the paper does not implement multimodal fusion layers or cross-modal attention mechanisms. Instead, it uses LLMs to process structured data converted into textual format, enabling anomaly detection through language-based reasoning. The study concludes that LLMs, especially when enhanced with HITL and task-oriented dialogues, offer a promising direction for improving cybersecurity in smart grid applications.

4.3. Semantic sensor interpretation

Combining sensor data with other inputs such as numerical sensor readings, metadata, historical trends, and equipment specs, including images derived either from documentation, natural language or other devices is a strength point of forecasting. Especially in the area of PV forecasting, as outlined in [34], the rapid advances in the emerging field of computer vision-based solar forecasting with deep learning motivate a renewed interest in the multimodal approach to PV output forecasting. The work in [35] already proposes a specialized multimodal transformer (the Auto Cross Modal Correlation Attention, ACMCA) tailored for PV forecasting. In this case, data fusion is performed at an early stage at the features level, as detailed in Fig. 11. The proposed architecture goes beyond standard multimodal transformers by introducing frequency-domain autocorrelation for time-series modeling. It also includes a directional cross-modal attention mechanism for deep fusion. Finally, the vision Transformer is also adapted for temporal image sequences. Similarly, an early fusion approach is used in [28] where a GPT-3.5 agent multimodal PV forecasting paper performs fusion at the feature-level, because it converts linguistic weather descriptions into numeric likelihoods and concatenates them directly with meteorological variables before modeling. In this case, the authors do not employ a multimodal LLM for forecasting itself; rather, they use GPT-3.5 merely as a pre-processing tool to interpret text descriptions, while the predictive core relies on two hierarchical XGBoost [105] models. In this way, the approach turns out to be not too computationally expensive, as the GPT-Agent method is lightweight because GPT is only queried once per weather description. The forecasting model itself, instead, is based on XGBoost, which is trained in only three minutes and runs in about one second including API calls. In [33], a Temporal Fusion Transformer, MATNet, receives historical PV measurements along with both historical and forecasted weather data. Since historical and forecasted weather inputs represent conditions observed at different points in time, they are inherently distinct. To account for this, MATNet adopts a multi-level joint fusion strategy in which historical PV, historical weather, and forecasted weather data are first embedded and processed separately, and only later fused through multiple abstraction layers inside its transformer architecture. Regarding model choice, MATNet, uses a fully deep-learning, transformer-based multimodal model that jointly learns from multiple time-series sources but does not incorporate an LLM. MATNet is significantly computationally heavier, as it uses high-dimensional embeddings, multi-head self-attention, dense interpolation layers, and multi-level fusion, and requires GPU-based training for 200 epochs with a 512-dimensional model and 8 attention heads, making it computationally intensive.

5. Conclusion

This paper has presented a comprehensive analysis of multimodal LLMs and their transformative potential for smart grid management and control. Our review demonstrates that the integration of diverse data modalities, including textual logs, sensor measurements, weather data, and equipment imagery, through advanced transformer architectures represents a paradigm shift from traditional rule-based systems to intelligent, adaptive grid management.

The key findings of this study highlight several critical advantages of multimodal LLMs over conventional approaches. First, their ability to process heterogeneous data streams simultaneously enables more contextual and accurate decision-making in complex scenarios such as fault diagnosis and emergency response. Second, the attention-based fusion mechanisms allow for selective integration of complementary information while suppressing redundant signals across modalities. Third, the natural language processing capabilities facilitate intuitive human-machine interfaces, enabling grid operators to interact with systems using plain language queries and receive interpretable explanations.

Our analysis of specific applications reveals that multimodal LLMs excel particularly in scenarios requiring the synthesis of structured and unstructured data. In fault detection and diagnosis, the combination of sensor telemetry with maintenance logs and equipment images significantly improves accuracy and reduces diagnostic time. For cybersecurity applications, the integration of network traffic data with threat intelligence reports enables more robust detection of sophisticated multi-stage attacks. In operational forecasting, the fusion of historical data with weather patterns and textual advisories enhances prediction reliability under uncertain conditions.

The evolution from early fusion strategies—which concatenate features at input level—to sophisticated mid and late fusion approaches using cross-attention mechanisms represents a significant advancement in handling the temporal and spatial characteristics inherent in power system data. These architectural innovations address the fundamental challenge of preserving modality-specific information while enabling meaningful cross-modal interactions.

Looking forward, several research directions emerge as particularly promising.

The development of domain-specific foundation models pre-trained on power system data could further enhance performance while reducing computational requirements. Physics-informed architectures that embed power system constraints and laws directly into the model structure offer potential for more reliable and interpretable solutions.

A promising research direction involves the integration of potentially interpretable forecasting models, also based on multimodal features, such as the Temporal Fusion Transformers, within management and control frameworks, thus enabling grid operators to act optimally under uncertain scenarios while providing interpretable decision rationales. Subsequent investigations may also assess the resilience of the developed strategies under measurement inaccuracies and unforeseen operational scenarios.

Additionally, the integration of real-time learning mechanisms and human-in-the-loop feedback systems could enable continuous adaptation to evolving grid conditions and operator preferences.

The transition toward intelligent grid management powered by multimodal LLMs represents not just a technological advancement but a fundamental shift toward more resilient, efficient, and sustainable power systems.

However, further work should also address the trade-off between computational complexity and performance, especially in large-scale, highly dynamic environments where real-time decision-making is critical.

As renewable energy integration continues to accelerate and grid complexity increases, the capacity of these models to process diverse information streams and provide contextual intelligence will become increasingly critical for maintaining grid stability while achieving decarbonization objectives. The frameworks and insights presented in this review provide a foundation for researchers and practitioners to harness these powerful technologies in addressing the challenges of modern smart grid operations.

CRediT authorship contribution statement

G. Cirrincione: Writing – original draft, Supervision, Conceptualization. **M. Di Silvestre:** Writing – review & editing. **R. Musca:** Writing – review & editing, Validation, Investigation, Formal analysis. **Z. Naeem:** Writing – original draft, Conceptualization. **E. Riva Sanseverino:** Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **G. Sciumè:** Writing – review & editing, Formal analysis. **G. Zizzo:** Writing – review & editing, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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